

Worksheet No. - 04

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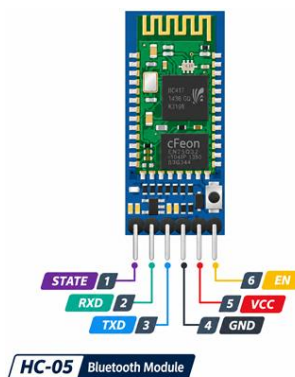
Aim/Overview of the practical: To interface an HC-05 Bluetooth module and an I2C LCD display with Arduino Uno, and to send text messages from an Android smartphone to the Arduino via Bluetooth so that the received message is displayed on the LCD screen.

APPARATUS / COMPONENTS REQUIRED

S.No.	Component	Quantity
1	Arduino Uno	1
2	HC-05 Bluetooth Module	1
3	16x2 LCD Display with I2C Adapter Module	1
4	Android Smartphone (with Bluetooth Serial Terminal app)	1
5	Breadboard	1
6	Jumper Wires (Male-Male / Male-Female)	As required
7	Resistors (1k Ω , 2k Ω) for voltage divider on HC-05 RXD	1 each
8	USB Cable (Type A to Type B)	1
9	5V Power Supply / USB Power Bank (optional)	1

THEORY

The HC-05 is a Serial Port Protocol (SPP) Bluetooth module that enables wireless serial communication between a microcontroller and a Bluetooth-enabled device such as a smartphone. The HC-05 Bluetooth module is a compact wireless communication device that enables data exchange between microcontrollers, computers, smartphones, and other Bluetooth-enabled devices using serial communication (UART). It is based on Bluetooth 2.0 + EDR technology and is commonly used in electronics projects, embedded systems, robotics, home automation, and wireless monitoring applications.



The HC-05 is popular because it is affordable, easy to configure, and compatible with popular development boards such as Arduino, ESP32, STM32, and Raspberry Pi. Its main purpose is to replace wired serial connections with a wireless Bluetooth link, allowing devices to transmit and receive data over short distances without requiring physical cables.

HC-05 Bluetooth Module Pinout and Pin Functions

Pin	Function	Description
STATE	Connection Status Output	The STATE pin indicates the current Bluetooth connection status of the module. It remains LOW when the HC-05 is not connected to another Bluetooth device and changes to HIGH after a successful Bluetooth connection is established.
RXD	Serial Data Input	The RXD pin receives serial data from a microcontroller or other host device. It should be connected to the TX (Transmit) pin of the microcontroller. Since the HC-05 Bluetooth chip operates at 3.3V logic, a voltage divider or level shifter is recommended when interfacing with 5V microcontrollers.
TXD	Serial Data Output	The TXD pin transmits serial data from the HC-05 to the connected microcontroller. It should be connected to the RX (Receive) pin of the microcontroller to enable two-way serial communication.
GND	Ground	The GND pin provides the common electrical reference for the module. It must be connected to the ground of the power supply and any connected microcontroller to ensure stable operation.
VCC	Power Supply Input	The VCC pin supplies power to the module. Most HC-05 breakout boards include an onboard voltage regulator, allowing them to accept an input voltage of approximately 3.6V to 6V, while internally powering the Bluetooth chip at 3.3V.
EN (KEY)	Enable / AT Mode Control	The EN pin is used to enable special operating modes, including AT command mode for configuration. On many HC-05 boards, pulling this pin HIGH during startup allows access to advanced settings such as the device name, baud rate, password, and operating mode.

It operates as a UART-to-Bluetooth bridge: any data sent from the paired Android device is received on the module's TXD pin and forwarded to the Arduino, and vice versa.

An I2C LCD module combines a standard 16x2 character LCD with a PCF8574 I/O expander chip, allowing the display to be controlled using only two wires (SDA and SCL) instead of the many parallel data lines a normal LCD requires. This greatly simplifies wiring and frees up Arduino pins.

CIRCUIT DIAGRAM / PIN CONNECTIONS

Connect the HC-05 module and the I2C LCD to the Arduino Uno as shown below. Note that the HC-05 RXD pin operates at 3.3V logic, so a voltage divider (1kΩ in series, 2kΩ to GND) must be used between Arduino TX (pin 1) and HC-05 RXD to avoid damaging the module.

Component	Component Pin	Arduino Uno Pin	Notes
HC-05 Bluetooth	VCC	5V	Power supply
	GND	GND	Ground connection
	TXD	Pin 0 (RX)	Transmits data to Arduino RX
	RXD	Pin 1 (TX)	Use voltage divider (Bluetooth RX is 3.3V logic)
I2C LCD (16x2)	VCC	5V	Power supply
	GND	GND	Ground connection
	SDA	A4 (or SDA pin)	I2C Data line
	SCL	A5 (or SCL pin)	I2C Clock line

PROGRAM / CODE

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <SoftwareSerial.h>

SoftwareSerial Bluetooth(10, 11);
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
    Serial.begin(9600);
    Bluetooth.begin(9600);

    lcd.init();
    lcd.backlight();

    lcd.setCursor(0, 0);
    lcd.print("Sukhleen Kaur");

    lcd.setCursor(0, 1);
    lcd.print("25MCA20031");

    delay(3000);
    lcd.clear();
}

void loop() {

    if (Bluetooth.available()) {
        String msg = Bluetooth.readStringUntil('\n');
        msg.trim();

        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Phone:");

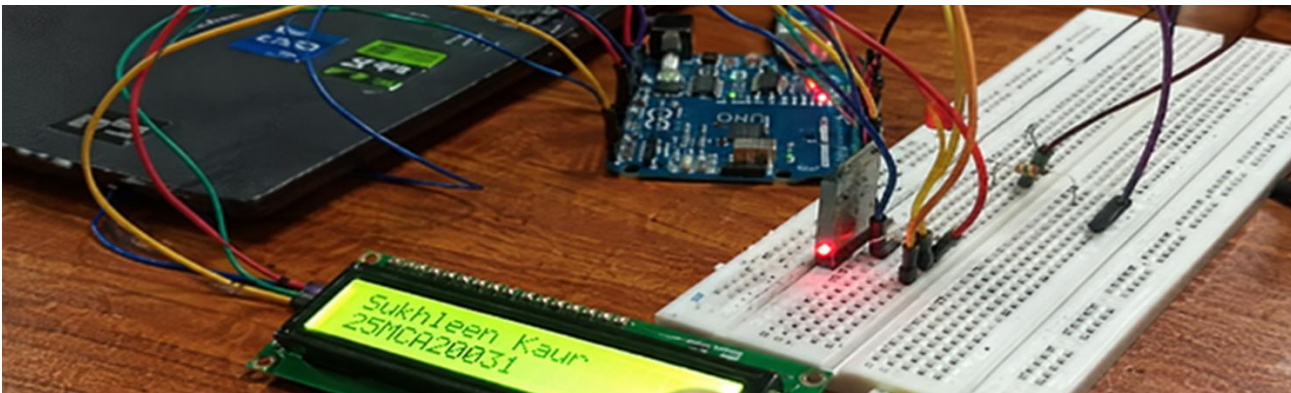
        lcd.setCursor(0, 1);
        lcd.print(msg.substring(0, 16));
    }

    if (Serial.available()) {
        String msg = Serial.readStringUntil('\n');
        msg.trim();

        Bluetooth.println(msg);
    }
}
```

```
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("Arduino:");  
  
lcd.setCursor(0, 1);  
lcd.print(msg.substring(0, 16));  
}  
}
```

OUTPUT



PROCEDURE

1. Wire the HC-05 module and I2C LCD to the Arduino Uno as per the connection table above.
2. Install the LiquidCrystal_I2C library in the Arduino IDE if not already installed.
3. Connect the Arduino Uno to the computer via USB and open the Arduino IDE.
4. Type or paste the program code into a new sketch.
5. Select the correct board (Arduino Uno) and COM port from the Tools menu, then compile and upload the code.
6. Power on the HC-05 module; its LED should blink rapidly, indicating it is ready to pair.
7. On the Android phone, open Bluetooth settings and pair with the HC-05 (default PIN: 1234 or 0000).
8. Open a Bluetooth Serial Terminal app on the phone and connect to the paired HC-05 device.
9. Type a text message in the app and send it.
10. Observe the message being displayed on the I2C LCD screen connected to the Arduino.

RESULT

Text messages sent from the Android smartphone were successfully transmitted via the HC-05 Bluetooth module and displayed in real time on the I2C LCD connected to the Arduino Uno, thereby demonstrating wireless serial communication between a mobile device and a microcontroller.

PRECAUTIONS

1. Ensure the HC-05 RXD pin receives only 3.3V logic; use a voltage divider or logic-level converter to avoid damaging the module.
2. Do not use Arduino pins 0 and 1 for SoftwareSerial, as they are shared with the USB programming interface.
3. Verify the I2C address of the LCD module (commonly 0x27 or 0x3F) using an I2C scanner sketch if the display does not initialize.
4. Keep all wiring connections secure to avoid intermittent communication or noisy/garbled data.
5. Ensure the Bluetooth module and LCD share a common ground with the Arduino.